



Identity and Access Management (IAM)

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Introduce

- AWS Identity and Access Management (IAM) enables to manage access to AWS services and resources securely.
- Create and manage AWS users and groups, and use permissions to allow and deny their access to AWS resources.



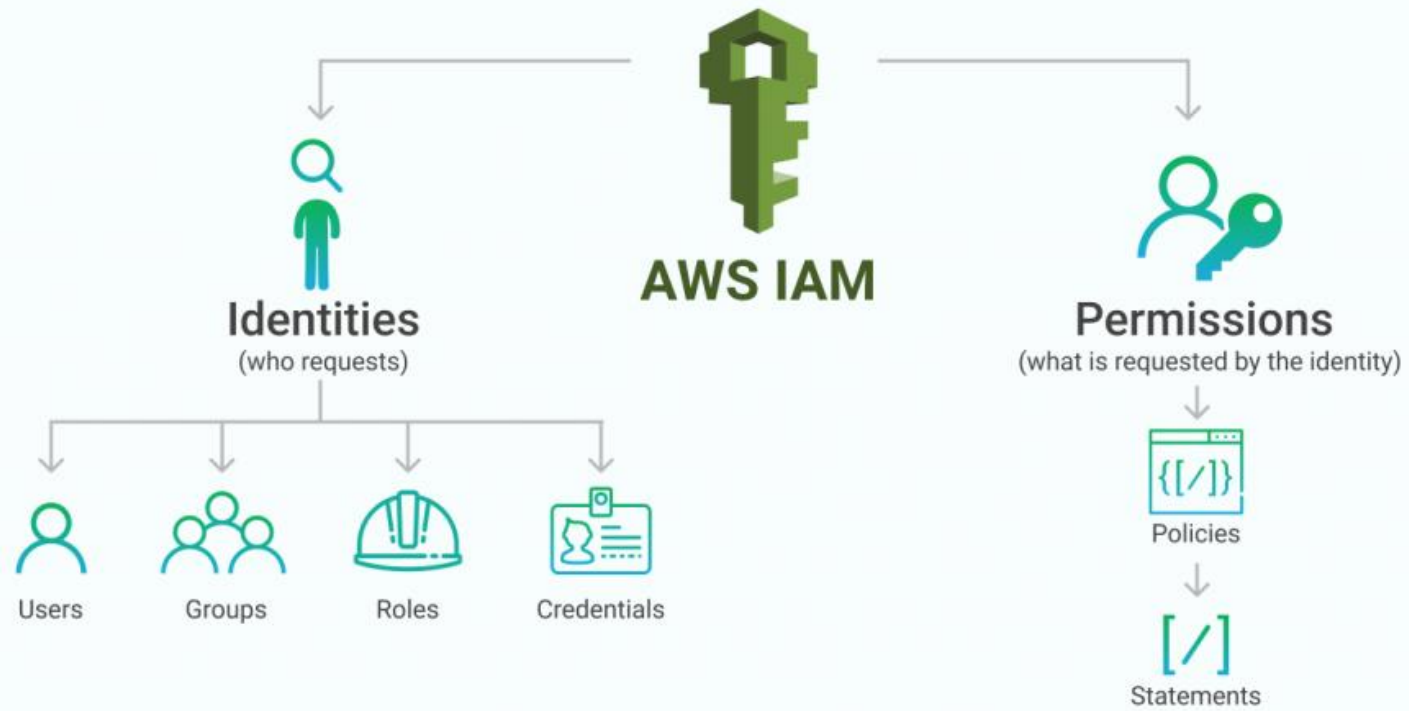
AWS IAM

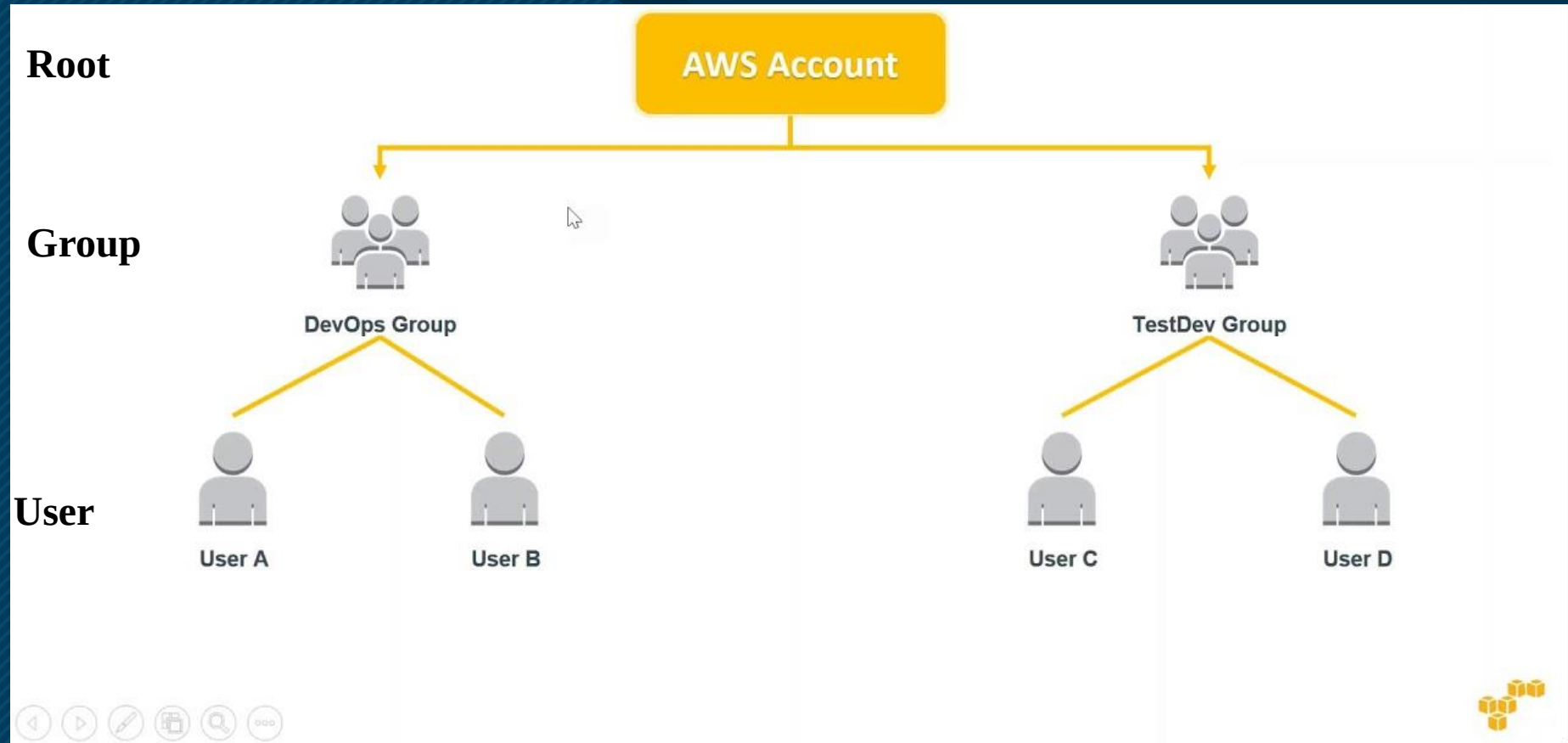
Use case:

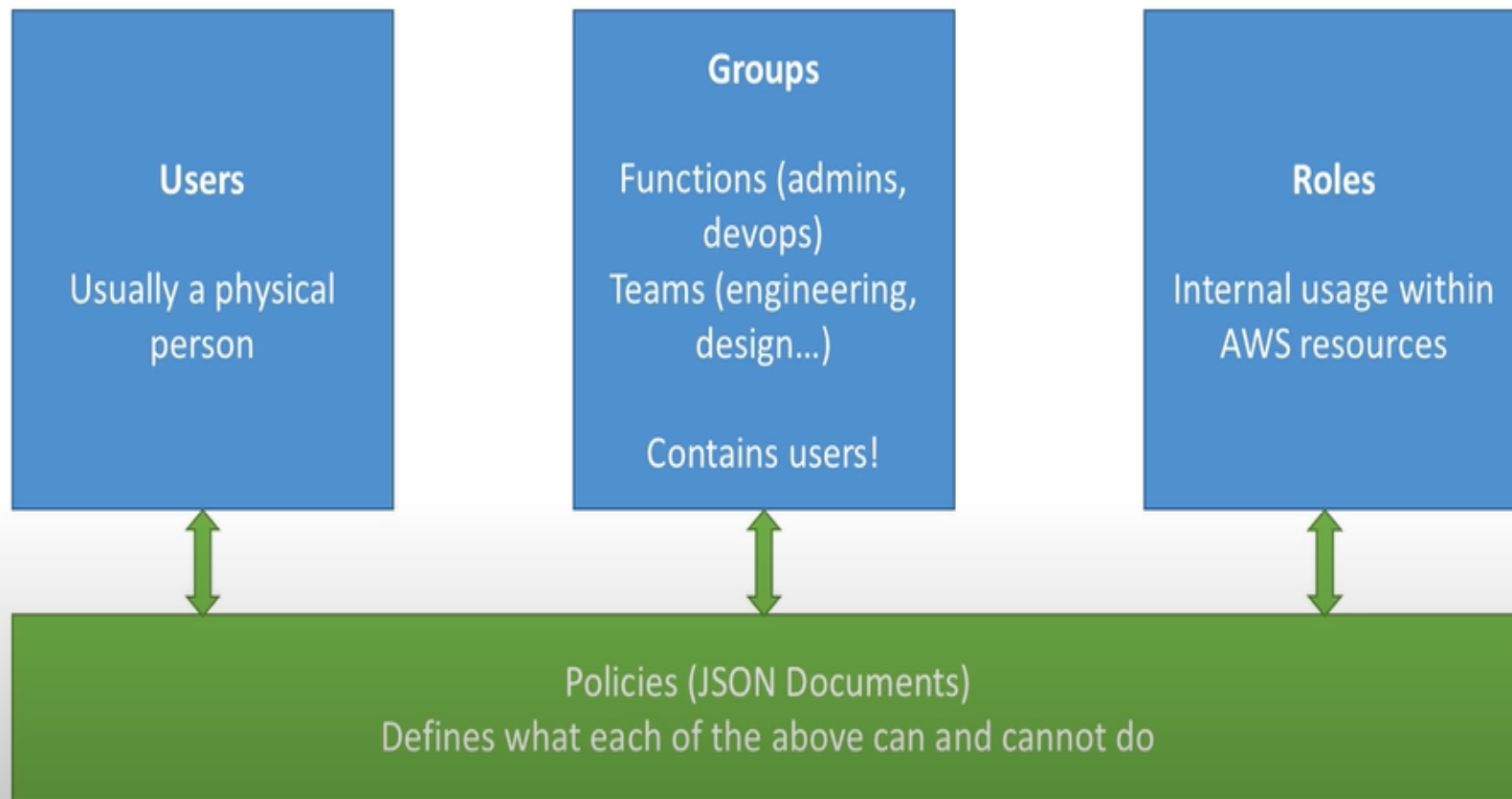
- Fine-grained access control to AWS resources
- Multi-factor authentication for highly privileged users
- Analyze access

IAM allows to:

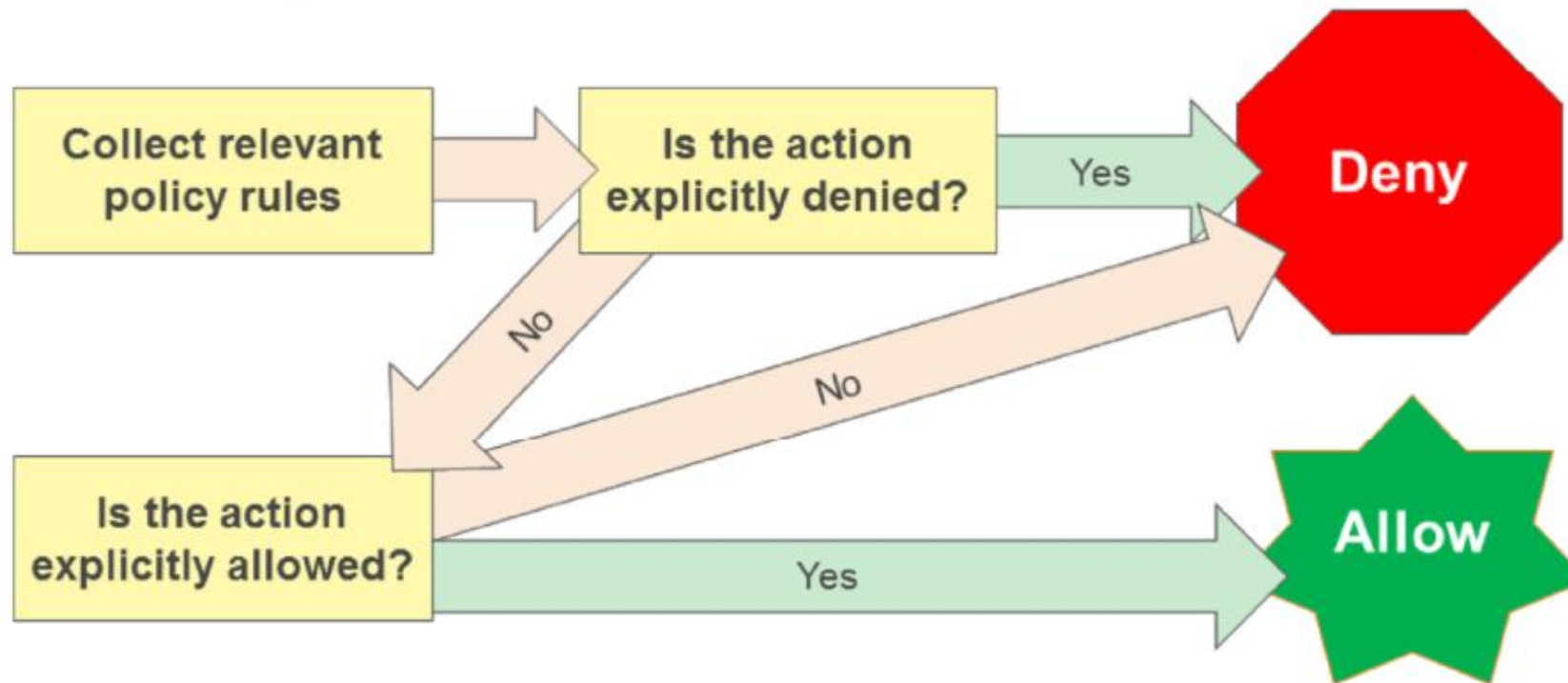
- Manage IAM users and their access
- Manage IAM roles and their permissions
- Manage federated users and their permissions

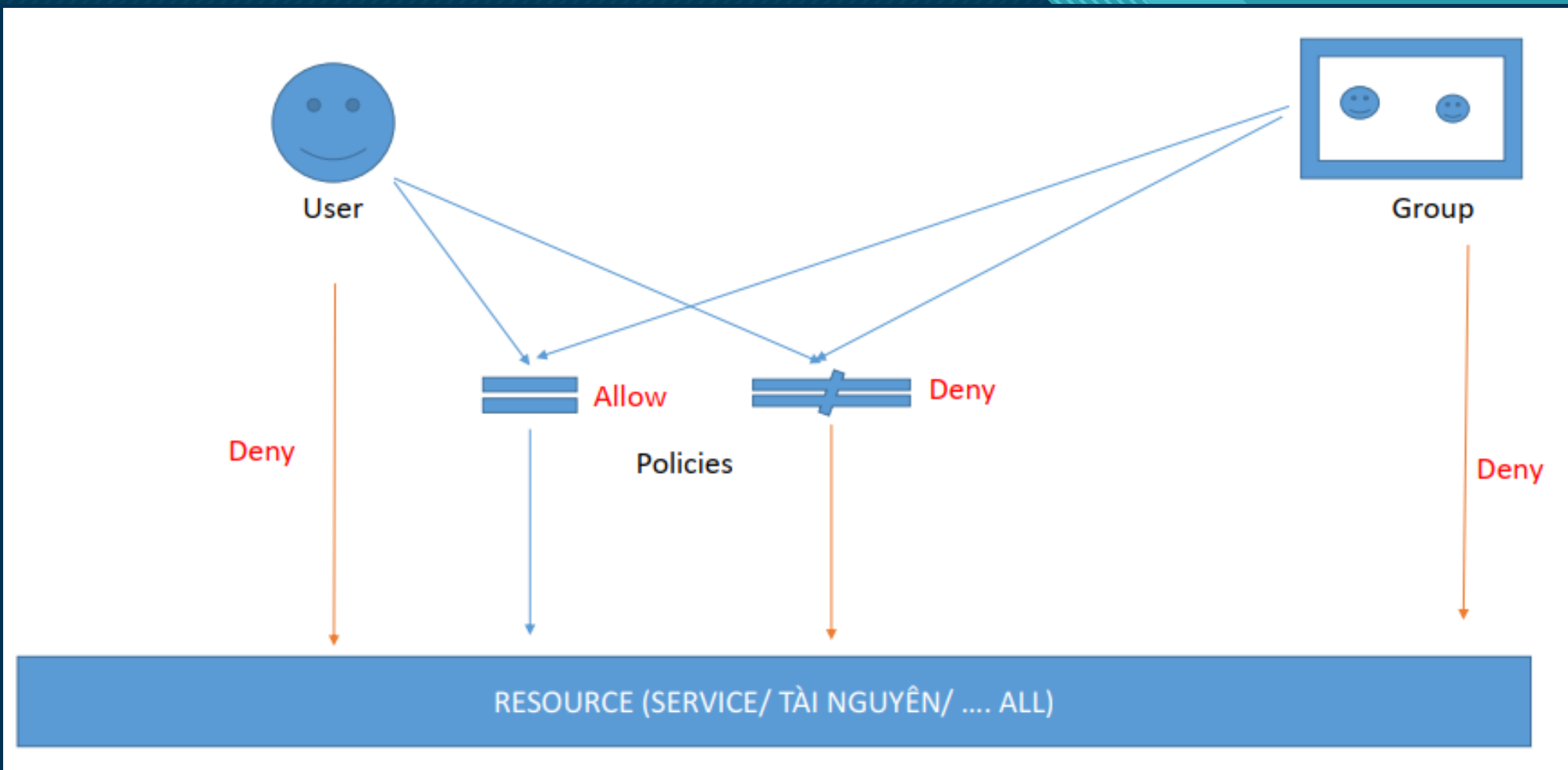




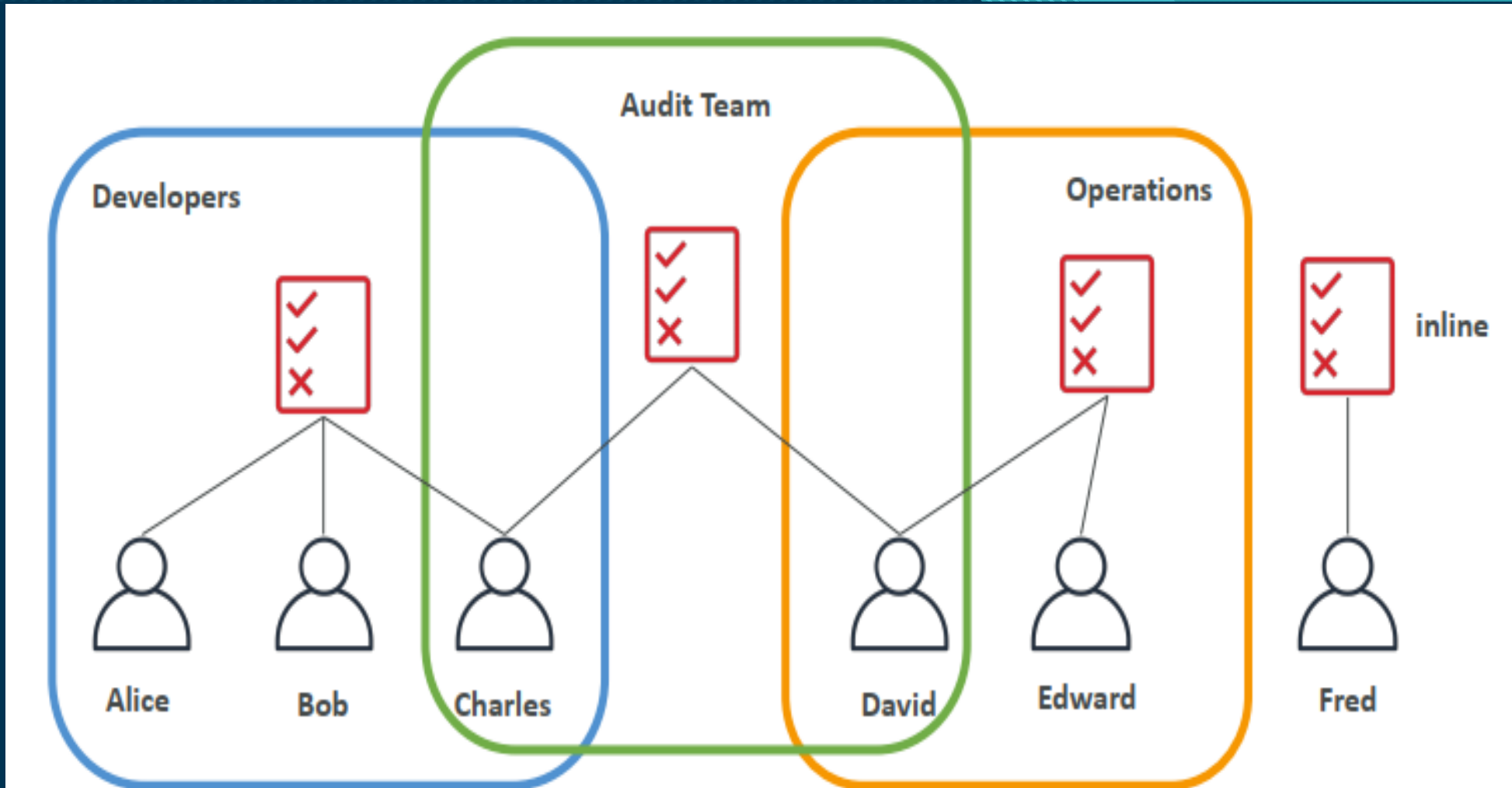


IAM Policy Rule Precedence





IAM Policies inheritance



IAM Policies Structure

Version: policy language version, always include "2012-10-17"

Sid: an identifier for the statement (optional)

Id: an identifier for the policy (optional)

Statement: one or more individual statements (required)

Resource: list of resources to which the actions applied to

```
{
  "Version": "2012-10-17",
  "Id": "S3-Account-Permissions",
  "Statement": [
    {
      "Sid": "1",
      "Effect": "Allow",
      "Principal": {
        "AWS": ["arn:aws:iam::123456789012:root"]
      },
      "Action": [
        "s3:GetObject",
        "s3:PutObject"
      ],
      "Resource": ["arn:aws:s3:::mybucket/*"]
    }
  ]
}
```

Effect: whether the statement allows or denies access (allow, deny)

Principal: account/user/role which this policy applied to

Action: list of actions this policy allows or denies

Condition: conditions for when this policy is in effect (optional)

- Password Policy
- Multi Factor Authentication (MFA)
- IAM security tool
- IAM Guidelines



Thank You!!!